ME551 Vibrations in Linear Systems

Spring Semester, 2024 **Department of Mechanical Engineering, KAIST**

Midterm Exam

1. (30) ***Fundamentals: True or False questions***

Answer to the **True or False** questions given below and explain briefly why. Without proper reasoning, no credit will be given even to the correct answers.

(a) (Linear Systems) Linear systems hold principle of superposition, i.e., both homogeneity and additivity, but nonlinear systems do not hold at least one of homogeneity and additivity. (5pts)

True. Any system which does not hold superposition is nonlinear system.

(b) (Convolution Theorem) Vibration response to an arbitrary excitation can be obtained by either convolution integral of impulse response function and input in the time domain; or convolution integral of frequency response function (FRF) with input in the frequency domain. (5pts)

False: It can be obtained by the multiplication not convolution integral of frequency response function (FRF) with input in the frequency domain.

(c) (Principle of Virtual Work) Virtual work is work done by the external force with arbitrary displacement from the equilibrium state compatible with geometric constraints. (5pts)

True: Virtual work is work done by the external force with virtual displacement

(d) (Harmonic Responses) For a harmonic input with frequency ω, we have a harmonic output with frequency ω for either linear or nonlinear systems. (5pts)

False: It is valid only for linear system. For nonlinear system we have different frequency terms in general depending on the type of nonlinearity

(e) (Stability) If there exists a negative definite function along the response trajectory whose total time derivative is positive definite, then the equilibrium state is asymptotically stable. (5pts)

True: From the Liapunov Stability Theorem II.

(f) (State Space Representation) Any p-th order linear MIMO systems with n-degrees of freedom can be formulated into a 1-st order state equation with p-degrees of freedom. (5pts)

False: It can be formulated into a 1-st order state equation with pn-degrees of freedom

2. (15) *1st and 2nd Order Systems, Frequency Response Functions*

Motion of a third order system is described by a differential equation as follows:

$$\frac{d^3x}{dt^3}+1.2\frac{d^2x}{dt^2}+4.2\frac{dx}{dt}+4x(t)=f(t).$$

(a) Decompose the frequency response function of the above third order system ($G_3(j\omega)$) into a linear superposition of two sub-systems, *i.e.*, one of the first order ($G_1(j\omega)$) and the other of the second order ($G_2(j\omega)$). (5pts)

Solution

Apply Laplace transform with zero initial condition.

$$(s^3+1.2s^2+4.2s+4)X=F$$

$$G_3(s)$$

$$=\frac{1}{s^3+1.2s^2+4.2s+4}$$

$$=\frac{1}{(s+1)(s^2+0.2s+4)}$$

$$=\frac{0.2083}{s+1}+\frac{-0.2083s+0.1667}{s^2+0.2s+4}$$

$$\Rightarrow G_1(j\omega)=\frac{0.2083}{j\omega+1},\quad G_2(j\omega)=\frac{-0.2083(j\omega)+0.1667}{-\omega^2+0.2(j\omega)+4}$$

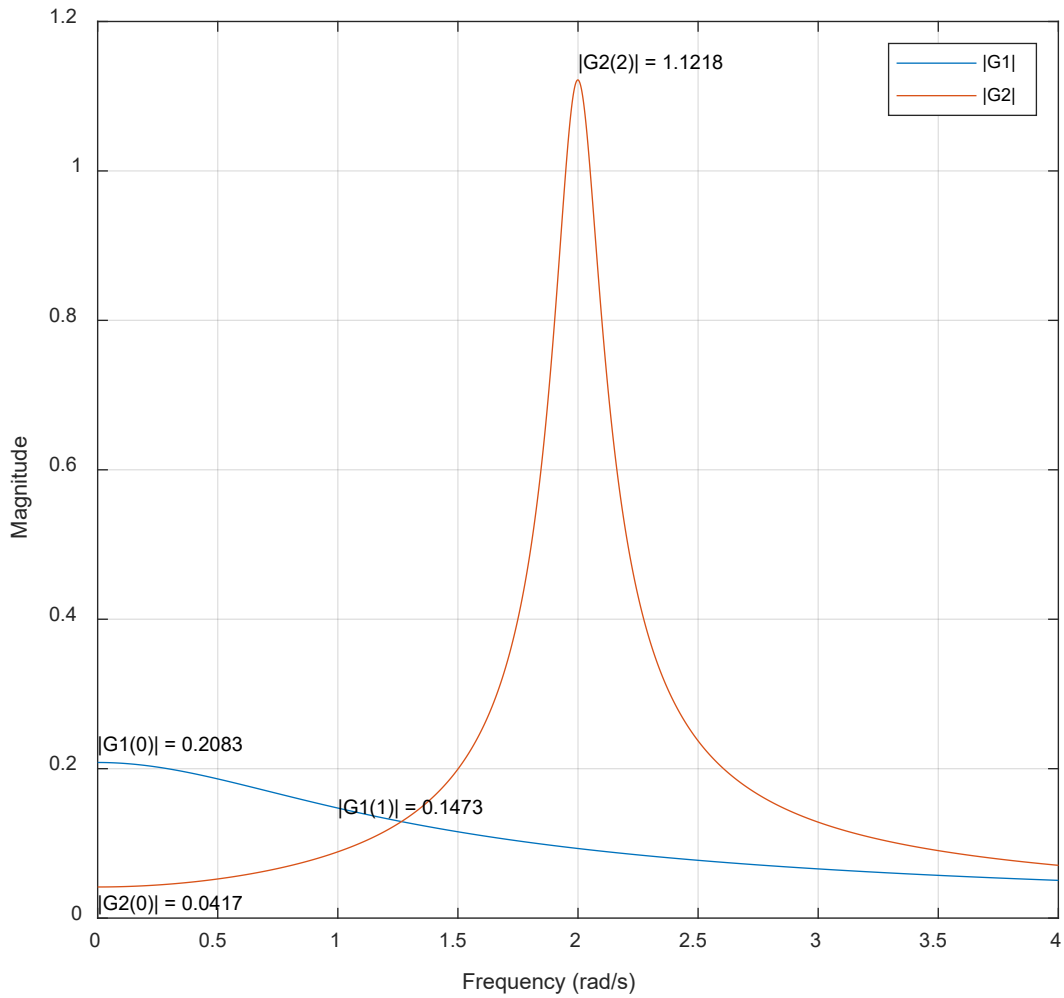
(b) A first order system is characterized typically by break frequency (ω_b) (or time constant) and an under-damped second order system is characterized typically by undamped natural frequency (ω_n) and damping ratio (ζ). Find out ω_b, ω_n and ζ. (5pts)

Solution

$\omega_b=1\,\text{rad/s},\ \omega_n=2\,\text{rad/s},\ \zeta=0.05$

(c) Sketch approximately the magnitudes of the frequency response functions of two sub-systems, putting coordinates of the characteristic points of each frequency response function such as $|G_1(0)|$, $|G_1(\omega_b)|$, $|G_1(\infty)|$, and $|G_2(0)|$, $|G_2(\omega_n)|$, $|G_2(\infty)|$ on the drawing. (5pts)

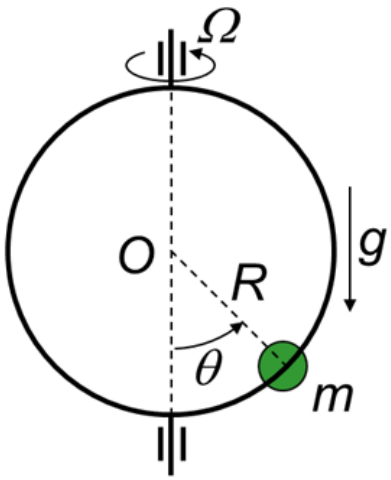
Solution



$$|G_1(\infty)| = |G_2(\infty)| = 0$$

3. (30) *Lagrange Equation, Equilibrium Point, Linearization, Stability*

The system in Fig. P3 consists of a smooth massless hoop of radius R rotating with a constant speed Ω about a vertical axis. A bead of mass m can slide freely along the hoop without friction. The angular position of the bead on the hoop is described by θ and the gravity is given by g.



(a) Derive equation of motion of mass m with respect to θ using the Lagrange equation. (10pts)

Solution

$$\tilde{r} = \mathrm{R}\tilde{\mathrm{e}}_r$$

$$\dot{\tilde{r}} = \mathrm{R}\,\tilde{w}\times\tilde{\mathrm{e}}_r \quad \left(\tilde{w} = \Omega\tilde{k} + \dot{\theta}\tilde{e}_\phi = -\Omega\cos\theta\tilde{e}_r + \Omega\sin\theta\tilde{e}_\theta + \dot{\theta}\tilde{e}_\phi\right)$$

$$= R\left(-\Omega\sin\theta\tilde{e}_\phi + \dot{\theta}\tilde{e}_\theta\right)$$

$$T = \frac{1}{2}m\dot{\tilde{r}}^2 = \frac{1}{2}mR^2\left(\Omega^2\sin^2\theta + \dot{\theta}^2\right)$$

$$V = mgR\left(1-\cos\theta\right)$$

$$L = \frac{1}{2}mR^2\left(\Omega^2\sin^2\theta + \dot{\theta}^2\right) - mgR\left(1-\cos\theta\right)$$

$$\frac{\partial L}{\partial\dot{\theta}} = \frac{1}{2}mR^2 2\dot{\theta} \quad ,\frac{\partial L}{\partial\theta} = \frac{1}{2}mR^2\Omega^2 2\sin\theta\cos\theta - mgR\sin\theta$$

$$\therefore \ddot{\theta} + \frac{g}{R}\sin\theta - \Omega^2\sin\theta\cos\theta = 0$$

(b) Find out equilibrium position(s) of the mass denoted by θ_0 and the required relationship between g, R and Ω which enables the mass to be in equilibrium for $0 < \theta_0 < \pi$. (5pts)

Solution

$$\ddot{\theta} + \frac{g}{R}\sin\theta - \Omega^2 \sin\theta\cos\theta = 0$$

$$\downarrow \dot{\theta} = \ddot{\theta} = 0 \;\; at \;\; \theta = \theta_0$$

$$\sin\theta_0\left(\frac{g}{R\Omega^2} - \cos\theta_0\right) = 0$$

$$\downarrow \sin\theta_0 \neq 0 \; (\theta_0 = 0 \;\; at \;\; \pi)$$

$$\cos\theta_0 = \frac{g}{R\Omega^2} < 1 \quad \rightarrow \quad \Omega^2 > \frac{g}{R}$$

(c) Linearize equation of motion for small motions around the equilibrium position θ_0 obtained in (b). (5pts)

Solution

$$\ddot{\theta} + \frac{g}{R}\sin\theta - \Omega^2 \sin\theta\cos\theta = 0$$

$$\downarrow \begin{cases} \theta = \theta_0 + \theta_1(t) \\ \cos\theta_0 = \dfrac{g}{R\Omega^2} \\ \theta_1 << \theta_0, \;\; \sin\theta_1 \approx \theta_1, \cos\theta_{11} \approx 1 \end{cases}$$

$$\ddot{\theta} + \left\{\frac{g}{R} - \Omega^2\left(\cos\theta_0\cos\theta_1 - \sin\theta_0\sin\theta_1\right)\right\}\left(\sin\theta_0\cos\theta_1 + \cos\theta_0\sin\theta_1\right) = 0$$

$$\approx \left(\frac{g}{R} - \Omega^2\cos\theta_0 + \Omega^2\sin\theta_0\,\theta_1\right)\left(\sin\theta_0 + \cos\theta_0\,\theta_1\right)$$

$$= \left(\cancel{\frac{g}{R} - \Omega^2\cos\theta_0}\right)\sin\theta_0 + \theta_1\left(\Omega^2\sin^2\theta_0 + \cos\theta_0\left(\cancel{\frac{g}{R} - \Omega^2\cos\theta_0}\right)\right)$$

$$\downarrow \cos\theta_0 = \frac{g}{R\Omega^2} < 1$$

$$\ddot{\theta}_1 + \Omega^2\sin^2\theta_0\,\theta_1(t) = 0$$

(d) Obtain natural frequency of the linearized system obtained in (c) in terms of g, R and Ω. (5pts)

Solution

$$\ddot{\theta}_1 + \Omega^2 \sin^2 \theta_0 \, \theta_1(t) = 0$$

$$\downarrow$$

$$\omega_n^{\ 2} = \Omega^2 \sin^2 \theta_0$$

$$\omega_n = \Omega \sin \theta_0 = \Omega \left\{ \frac{\sqrt{(R\Omega)^2 - g^2}}{R\Omega^2} \right\}$$

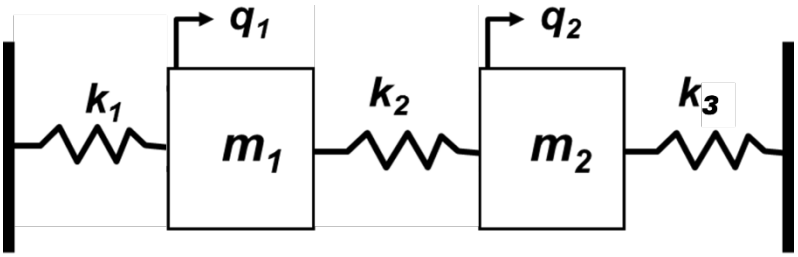
(e) Indicate the stability of the linearized system obtained in (c) by choosing, stable, asymptotically stable, and unstable and explain supporting reason for the stability characteristics. (5pts)

Solution

Stable vibration (Pure sinusoidal solution)

4. (25) ***Multiple DOF Systems, Eigenvalue Problems, Modal Analysis***

Consider the free vibration of an undamped two-DOF system shown in Fig. P4, where the mass and the stiffness are given by $m_1 = m_2 = 4\,\text{kg},\ k_1 = k_2 = k_3 = 16\,\text{N/m}$; and initial conditions $q_1(0) = q_2(0) = 1\,m,\ \dot{q}_1(0) = \dot{q}_2(0) = 0\,m/s$.



(a) Derive <u>matrix-vector equation of motion</u> of the free vibration $\boldsymbol{M}\ddot{\boldsymbol{q}} + \boldsymbol{K}\boldsymbol{q} = \boldsymbol{0}$ with Lagrange Equation. Drive <u>the corresponding symmetric eigenvalue problem</u> assuming harmonic solution of natural vibration such that $\boldsymbol{q}(t) = \boldsymbol{U}e^{j\omega t}$. (5pts)

Solution

$$T = \frac{1}{2}m_1\dot{q}_1^{\,2} + \frac{1}{2}m_2\dot{q}_2^{\,2}$$

$$V = \frac{1}{2}k_1q_1^{\,2} + \frac{1}{2}k_2(q_2 - q_1)^2 + \frac{1}{2}k_3q_2^{\,2}$$

$$L = T - V = \frac{1}{2}m_1\dot{q}_1^{\,2} + \frac{1}{2}m_2\dot{q}_2^{\,2} - \frac{1}{2}k_1q_1^{\,2} - \frac{1}{2}k_2(q_2 - q_1)^2 - \frac{1}{2}k_3q_2^{\,2}$$

1) $\dfrac{d}{dt}\left(\dfrac{\partial L}{\partial \dot{q}_1}\right) - \dfrac{\partial L}{\partial q_1} = 0$

$\Rightarrow m_1\ddot{q}_1 + (k_1 + k_2)q_1 - k_2q_2 = 0$

2) $\dfrac{d}{dt}\left(\dfrac{\partial L}{\partial \dot{q}_2}\right) - \dfrac{\partial L}{\partial q_2} = 0$

$\Rightarrow m_2\ddot{q}_2 + (k_2 + k_3)q_2 - k_2q_1 = 0$

The above equation can be considered as $M\ddot{q} + Kq = Q$, then the natural frequencies and mode shape vectors can be obtained by solving eigenvalue problem, $\omega^2 Mu = Ku$.

$$\mathbf{M}=\begin{bmatrix}4 & 0\\ 0 & 4\end{bmatrix},\mathbf{K}=\begin{bmatrix}32 & -16\\ -16 & 32\end{bmatrix}$$

$$\therefore \begin{bmatrix}4 & 0\\ 0 & 4\end{bmatrix}\ddot{\mathbf{q}}+\begin{bmatrix}32 & -16\\ -16 & 32\end{bmatrix}\mathbf{q}=\mathbf{0}$$

$$[\mathbf{U},\mathbf{D}]=eig(\mathbf{K},\mathbf{M})$$

$$\mathbf{U}=\begin{bmatrix}-0.3536 & -0.3536\\ -0.3536 & 0.3536\end{bmatrix}$$

$$\uparrow\mathbf{u}_1 \qquad \uparrow\mathbf{u}_2$$

$$\mathbf{D}=\begin{bmatrix}4 & 0\\ 0 & 12\end{bmatrix}\rightarrow \omega_1=2,\ \ \omega_2=2\sqrt{3}$$

(b) Construct <u>standard eigenvalue problem</u> $\boldsymbol{Av}=\lambda\boldsymbol{v}$ ($\lambda=\omega^2$) from coordinate transform of $\boldsymbol{u}=\boldsymbol{M}^{-1/2}\boldsymbol{v}$ and obtain its two <u>eigenvalues</u> λ_1,λ_2, and <u>corresponding eigenvectors</u> $\boldsymbol{v}_1,\boldsymbol{v}_2$.(5pts)

Solution

$$\mathbf{A}=\mathbf{M}^{-1/2}\mathbf{K}\,\mathbf{M}^{-1/2}=\begin{bmatrix}8 & -4\\ -4 & 8\end{bmatrix}$$

$$\mathbf{V}=\begin{bmatrix}-0.7071 & -0.7071\\ -0.7071 & 0.7071\end{bmatrix}$$

$$\uparrow\mathbf{v}_1 \qquad \uparrow\mathbf{v}_2$$

$$\rightarrow \lambda_1=4,\lambda_2=12$$

(c) Calculate <u>modeshapes</u> $\boldsymbol{U} = [\boldsymbol{u}_1, \boldsymbol{u}_2]$ and <u>confirm the orthogonality</u> of modeshape vectors such that $\boldsymbol{U}^T\boldsymbol{M}\boldsymbol{U} = \boldsymbol{I}$ and $\boldsymbol{U}^T\boldsymbol{K}\boldsymbol{U} = diag(\lambda_1, \lambda_2)$. (5pts)

Solution

$$\mathbf{u}_1 = \mathbf{M}^{-1/2}\mathbf{v}_1 = \begin{bmatrix} -0.3536 \\ -0.3536 \end{bmatrix}$$

$$\mathbf{u}_2 = \mathbf{M}^{-1/2}\mathbf{v}_2 = \begin{bmatrix} -0.3536 \\ 0.3536 \end{bmatrix}$$

$$\mathbf{U} = \begin{bmatrix} -0.3536 & -0.3536 \\ -0.3536 & 0.3536 \end{bmatrix}$$

$$\mathbf{U}^T\mathbf{M}\,\mathbf{U} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\mathbf{U}^T\mathbf{K}\,\mathbf{U} = \begin{bmatrix} 4 & 0 \\ 0 & 12 \end{bmatrix}$$

(d) Construct <u>decoupled free vibration equation</u> in terms of modal coordinates $\eta_1(t), \eta_2(t)$ using the modal expansion as $\boldsymbol{q}(t) = \boldsymbol{u}_1\eta_1(t) + \boldsymbol{u}_2\eta_2(t)$, and <u>obtain the modal response</u> $\eta_1(t), \eta_2(t)$ using the SDOF free vibration solution with an initial displacement η_0 and initial velocity $\dot{\eta}_0$ as $\eta(t) = \eta_0 \cos\omega_n t + \dfrac{\dot{\eta}_0}{\omega_n}\sin\omega_n t$. (5pts)

Solution

$$q_1(0) = q_2(0) = 1$$
$$\dot{q}_1(0) = \dot{q}_2(0) = 0$$

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathbf{0}$$
$$\downarrow \mathbf{q} = \mathbf{U}\boldsymbol{\eta}$$
$$\mathbf{I}\,\ddot{\boldsymbol{\eta}} + \begin{bmatrix} \omega_1^2 & \\ & \omega_2^2 \end{bmatrix}\boldsymbol{\eta} = 0$$

$$\mathbf{U}^T\mathbf{M}\mathbf{q}_0 = \mathbf{U}^T\mathbf{M}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2.8284 \\ 0 \end{bmatrix} = \boldsymbol{\eta}_0$$

$$\mathbf{U}^T\mathbf{M}\dot{\mathbf{q}}_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \dot{\boldsymbol{\eta}}_0$$

$$\therefore \eta_1(t) = -2.8284\cos 2t,\ \eta_2(t) = 0$$

(e) Obtain the <u>physical solution</u> $\boldsymbol{q}(t) = \boldsymbol{u}_1\eta_1(t) + \boldsymbol{u}_2\eta_2(t)$ from the modal solution $\eta_1(t), \eta_2(t)$ obtained in (d). (5pts)

Solution

$$\begin{bmatrix} q_1(t) \\ q_2(t) \end{bmatrix} = \begin{bmatrix} u_1 & u_2 \end{bmatrix} \begin{bmatrix} \eta_1(t) \\ \eta_2(t) \end{bmatrix}$$

$$= \begin{bmatrix} -0.3536 & -0.3536 \\ -0.3536 & 0.3536 \end{bmatrix} \begin{bmatrix} -2.8284 \\ 0 \end{bmatrix} \cos 2t$$

$$= \begin{bmatrix} \cos 2t \\ \cos 2t \end{bmatrix}$$

q_1 and q_2 have only the first mode.

Bonus Problem

(5pts) Please make a mechanical vibration problem by yourself and show the solution procedure. Any problem ideated from ME551 and proper answer will be fully credited.